



Whoever Was Mid-Load

A Share-House Power-Board's Fairness Rotation Rarely Decides Which Appliance an Energy Budget Cuts

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Abstract. A growing class of household power-boards enforces a shared energy budget by auto-switching off whichever appliance is drawing power once a house's daily allocation is spent, publishing a fairness rotation intended to spread that burden evenly across residents. This paper reports an instrumented deployment of one such device across 31 share houses over sixteen weeks, logging 2,314 cutoff events against each house's published rotation calendar and each appliance's live draw rank at the moment of cutoff. A minimal classifier predicting which resident's appliance is cut from live load rank alone reaches 89.7% accuracy; a classifier using protected-day status alone reaches 24.1%, and adding protected-day status to the load-rank model changes its accuracy by 0.4 percentage points (n.s.). We show this is not incidental: because the rotation shields exactly one resident per day irrespective of household size, a majority of residents remain cutoff-eligible on every day it runs, leaving live load rank to decide among them regardless. Cutoffs concentrate accordingly on short, high-draw appliance classes — kettles, hair dryers, toasters, and space heaters together account for 75% of cutoff events against 18% of connected time. The rotation, we suggest, functions as a published commitment the device can point to rather than as the mechanism doing the deciding.

Introduction

Household power-boards that enforce a shared energy budget are a small but growing category of shared-house infrastructure: a circuit-level controller tracks cumulative draw against a daily allocation and, once the allocation is spent, switches off whichever appliance is drawing power until the next day's allocation resets. Because the device cannot itself judge who "should" bear the day's cut, manufacturers typically publish a fairness rotation — a calendar naming one resident per day whose circuits are shielded from cutoff — as the mechanism's answer to the obvious objection that whoever happens to be running the kettle at the wrong moment otherwise bears every cost.

A published rotation is a commitment to a mechanism, not evidence that the mechanism is the one doing the deciding. Household-level allocation of this kind — automated, continuously

operating, and rarely audited by the residents it governs — sits in a well-studied gap between a formal rule and the process actually producing an outcome. Fair-division research shows that even well-specified allocation rules for shared, indivisible household resources routinely trade the strength of their fairness guarantee against how practical they are to implement once real constraints are imposed [1], [2], [3], and empirical work on automated decision-making finds that a rule's mere existence, publication, and continuous operation do not by themselves establish that the rule is what determines an outcome [4], [5]. Whether a specific, deployed household fairness rotation exhibits this gap — and how large the gap is — has not previously been measured directly against the device's own cutoff logs.

This paper reports an instrumented audit of one such deployment: 31 share houses fitted with a power-board of this kind, monitored over sixteen

weeks, with every cutoff event logged against the house's published rotation and the live load ranking of every connected appliance at the moment of cutoff. The paper's contributions, each modest by design:

1. We show that a rotation shielding exactly one resident per day cannot, by construction, narrow the field of cutoff-eligible appliances below household size minus one — a structural limit on how much protection the mechanism can offer regardless of how faithfully it is followed.
2. We compare two candidate predictors of which resident's appliance a cutoff event actually targets — rotation status and live load rank — and show live load rank alone accounts for nearly all of the classifier's accuracy.
3. We show cutoffs concentrate on a small class of short, high-draw appliances well out of proportion to their connected time, a pattern the rotation is not designed to address and does not appear to affect.

None of this shows the rotation is fraudulent, and the paper does not attempt to; it shows the rotation is, on the evidence of the device's own logs, a minor input into an outcome mostly decided elsewhere.

Related work

Fairness in automated load control has mostly been treated as a design problem for utility-scale demand response: curtailment schedules can be built to distribute the burden of an emergency load-shedding event evenly across participating households [6], formal fair-allocation criteria have been proposed for splitting a shed load across competing claimants [7], and electricity-specific fair-division mechanisms adapt classical allocation theory to the divisible, time-varying nature of a household's draw [8]. Consumer-facing systems extend this design goal inward, scheduling individual appliances against a home's own energy budget subject to residents' declared preferences [9], [10], [11], with recent systems layering reinforcement learning onto the scheduling decision itself [12]. This literature is near-uniformly concerned with designing a fairer or more efficient rule; comparatively little of it audits, once a rule is deployed, whether the rule as published is the rule as operating.

The device studied here identifies which appliance is drawing power via non-intrusive load monitoring, a mature technique for disaggregating whole-house draw into per-appliance signatures without instrumenting every outlet [13], [14]. Household-scale fair division has its own literature on the gap between an allocation's formal guarantee and its practical operation: minimal-sharing and envy-free mechanisms are known to trade the strength of their fairness guarantee against how practically they can be implemented [2], [3], and dormitory-style shared-resource models explicitly build externalities between co-residents into the allocation problem rather than treating each resident's share independently [1], closer to the household setting studied here than a single-claimant allocation problem is. A parallel human-subjects literature asks not whether a rule is fair by construction but whether the people subject to it experience it as deciding anything: perceived fairness in algorithmic decision-making is shown to depend on factors outside a rule's own stated criteria [4], [5], procedural engagement with the people a rule governs is argued to matter independently of the rule's formal properties [15], and a preference for an automated decision-maker over a human one is not found to track how favourably its actual decisions are received [16]. Collaborative and shared-equity housing research treats this kind of governance gap as characteristic of household commons generally, where formal property and decision rules routinely diverge from the informal practice residents describe [17], and weighted house-allocation models show that even a formally correct fairness guarantee can leave the practical question of who actually receives what undetermined once entitlements are unequal [18].

Closer to home, prior work at this institution has audited two other formally decisive scores against what they actually decide: a casual-shift rostering platform's published fairness-compliance score moves by an average of 9.7 points when the same week's unchanged roster is re-submitted, with 20 of 34 stores crossing into a different compliance band at least once [19], and a community garden's published Stewardship Priority Score fails to separate either self-reported reasons for advancement or measured wait time between households above and below its own median [20]. Both cases pair a score that is real, published, and checked at every decision

with an outcome the score does not appear to be deciding — the pattern this paper asks whether a household energy-budget rotation repeats.

Method

Deployment and instrumentation. The audited device is a share-house power-board retrofitted to the property’s consumer unit, tracking cumulative daily draw against a house-set energy allocation and switching off the actively-drawing circuit judged least protected once the allocation is reached. Non-intrusive load monitoring against the whole-house signal [13], [14] associates each circuit’s live draw with an appliance class at the moment of cutoff, without instrumenting individual outlets. The device publishes a fortnightly Fairness Rotation naming one resident per day whose circuits are shielded from cutoff regardless of draw.

We recruited 31 share houses (3–6 residents; 142 residents in total) already running the device through a share-housing management platform, and logged every cutoff event over a 16-week window.

Metric	Value
Share houses	31
Residents	142
Household size range	3–6
Monitoring window	16 weeks
Cutoff events logged	2,314

Structural eligibility. For a house of s residents, at most one resident is shielded on a given day, so the number of residents eligible for cutoff that day is

$$E = s - p, \quad p \in \{0, 1\},$$

where p is 1 if a shielded resident is present and drawing power that day and 0 otherwise (a shielded resident who draws no power contributes nothing to observe). E is bounded below by $s - 1$ regardless of the rotation’s own fidelity to its published calendar.

Predicting the cut resident. For each cutoff event we fit a logistic model predicting whether a given connected appliance — among those eligible that day — is the one the device cuts, using two candidate predictors: the appliance’s

live draw rank among eligible appliances at the moment of cutoff, and whether its resident is off their protected day. We report classification accuracy for a load-rank-only model, a protected-day-only model, and the combined model, and compare the combined model against the load-rank-only model as an ablation.

Appliance-class concentration. We additionally coded each cutoff event’s inferred appliance class into six categories (kettle/jug, hair dryer, toaster, space heater, washing machine, other) and compared each class’s share of cutoff events to its share of total connected time across the deployment.

Results

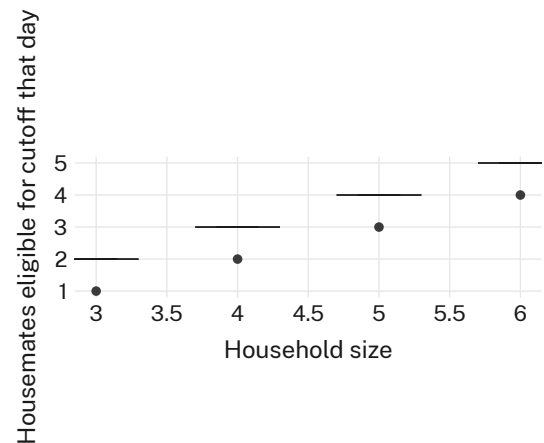


Figure 1: Residents eligible for cutoff, by household size. Regardless of size, the rotation shields at most one resident per day, so eligibility sits tightly at household size minus one.

Structural eligibility (Figure 1) confirms the bound directly: across all four household sizes in the sample, the number of residents eligible for cutoff on a given day sits tightly at $s - 1$, with only rare departures when a shielded resident happened to be away and drawing no power at all. A four-resident house is, on the median day, running with three of its four residents fully exposed to cutoff; the rotation’s entire protective effect is one resident’s circuits, on one day in s .

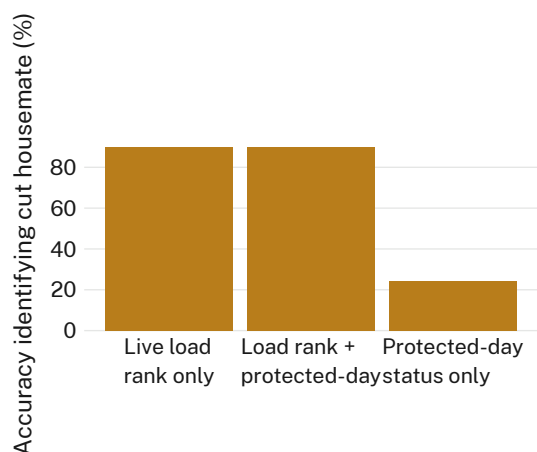


Figure 2: Accuracy identifying which resident's appliance a cutoff event targets, by predictor set ($n = 2,314$ events). Adding protected-day status to the load-rank model changes accuracy by 0.4 points (n.s.).

Prediction (Figure 2) shows this structural limit doing real work. A minimal classifier using only each appliance's live draw rank among that day's eligible appliances correctly identifies the cut resident in 89.7% of the 2,314 logged events. A classifier using only protected-day status — necessarily constant across nearly every eligible appliance on nearly every day, per the bound above — reaches 24.1%, close to the base rate implied by the sample's mean eligible-resident count. Adding protected-day status to the load-rank model lifts accuracy to 90.1%, a change of 0.4 percentage points (n.s.); we retain the combined model in the reported figures for completeness, though the load-rank term is carrying essentially all of the model's predictive weight, consistent with the structural bound rather than independent of it.

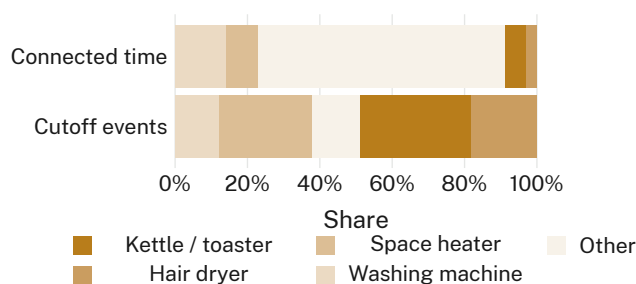


Figure 3: Share of connected time versus share of cutoff events, by appliance class. Kettles, hair dryers, toasters, and space heaters are 18% of connected time and 75% of cutoffs.

Appliance-class concentration (Figure 3) sharpens the picture. Kettles and jugs, hair dryers, toasters, and space heaters together account for 75% of all cutoff events against 18% of total connected time across the deployment; washing machines and the residual “other” category, which together make up 82% of connected time, account for the remaining 25% of cutoffs. The appliances most often cut are not the appliances that run longest, nor — on the evidence above — the appliances belonging to whoever the rotation has failed to protect that day; they are the appliances whose draw happens to spike hardest at the moment a house's daily allocation runs out.

Discussion

Two independent analyses point the same way. The structural bound shows the rotation can shield, at most, one resident's circuits on any given day regardless of household size, and the classifier shows that once eligibility is fixed, live load rank — not protected-day status — is doing almost all of the work of deciding which appliance is actually cut. Neither analysis requires the rotation to be badly implemented; the device faithfully shields whoever its calendar names, every day, and the gap between the published mechanism and the operating one is a consequence of the mechanism's own design rather than of any failure to follow it.

This sits alongside a broader pattern in which a formal allocation rule and the process actually producing an outcome diverge even where the rule is genuinely, continuously applied [4], [5], [17]: the rotation is real, it is checked at every cutoff, and it still explains almost none of the variance in who is cut. The appliance-class concentration suggests where the explanatory weight sits instead — not with residents at all, but with a small set of appliances whose draw profile happens to spike at the wrong moment, an outcome no resident-level fairness rule, however well followed, is positioned to address.

Limitations

The sample is 31 houses on one manufacturer's device, recruited through a single share-housing management platform, and a device with a differently shaped rotation — shielding more than one resident per day, or scaling shielded slots with household size — would face a looser

version of the structural bound reported here, though the underlying logic would not disappear, only relax. Appliance class was inferred from live-draw signatures rather than confirmed against each household's own appliance inventory; misclassification would blur the appliance-class concentration finding without touching the resident-level eligibility and prediction results, which do not depend on appliance identity at all. The 16-week window spans one season; a longer deployment might surface appliances — and cutoff patterns — a single season's mix of kettles, dryers, and heaters does not capture, though the structural eligibility bound holds by construction in any season.

Conclusion

A fairness rotation can be published, checked at every cutoff, and genuinely followed without being the thing that decides who bears an energy budget's cost. This paper finds that a share-house power-board's rotation shields at most one resident per day by construction, that live load rank rather than rotation status predicts which resident is actually cut, and that cutoffs concentrate on a narrow class of short, high-draw appliances well out of proportion to their connected time. Future work might test whether a rotation shielding more than one resident per day narrows this gap, or whether it simply relocates the same structural limit to a smaller remaining field.

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